

CS100433

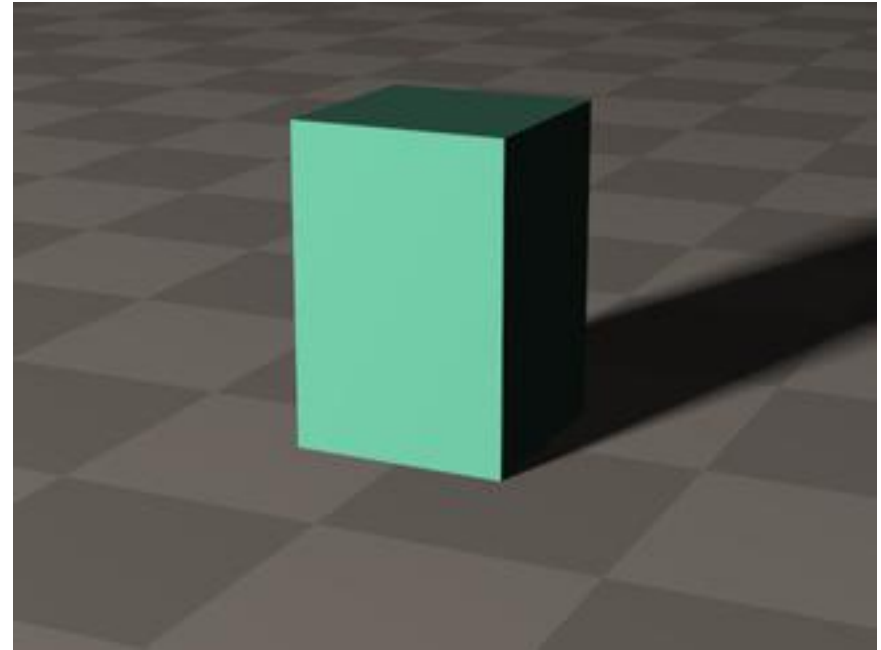
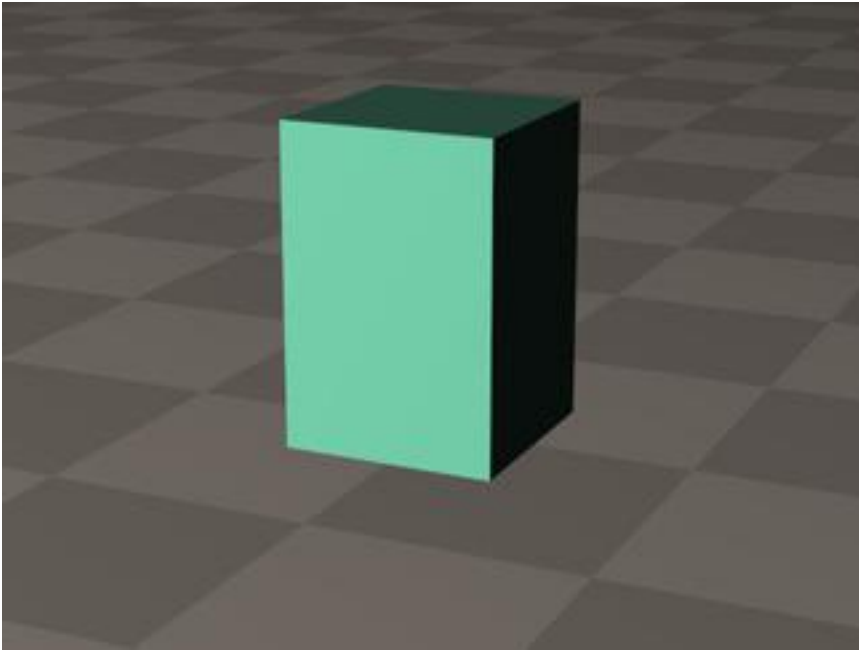
Shadows

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Why shadows?

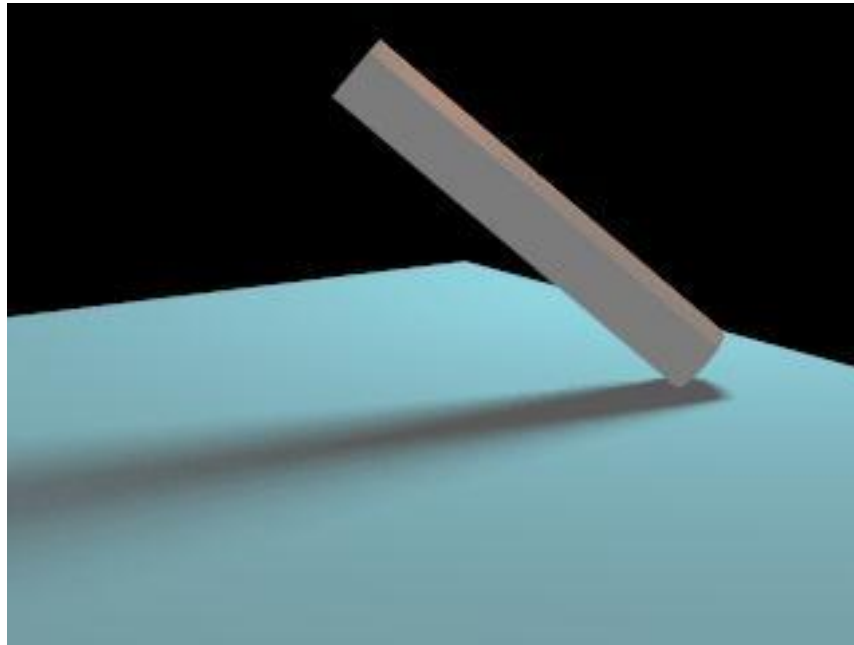
- 3D cueing
 - location, even motion



[Tom Thorne, Edinburgh]

How shadows are generated?

- Areas hidden from the light source by occlusion cause by objects
- Hard shadows and soft shadows



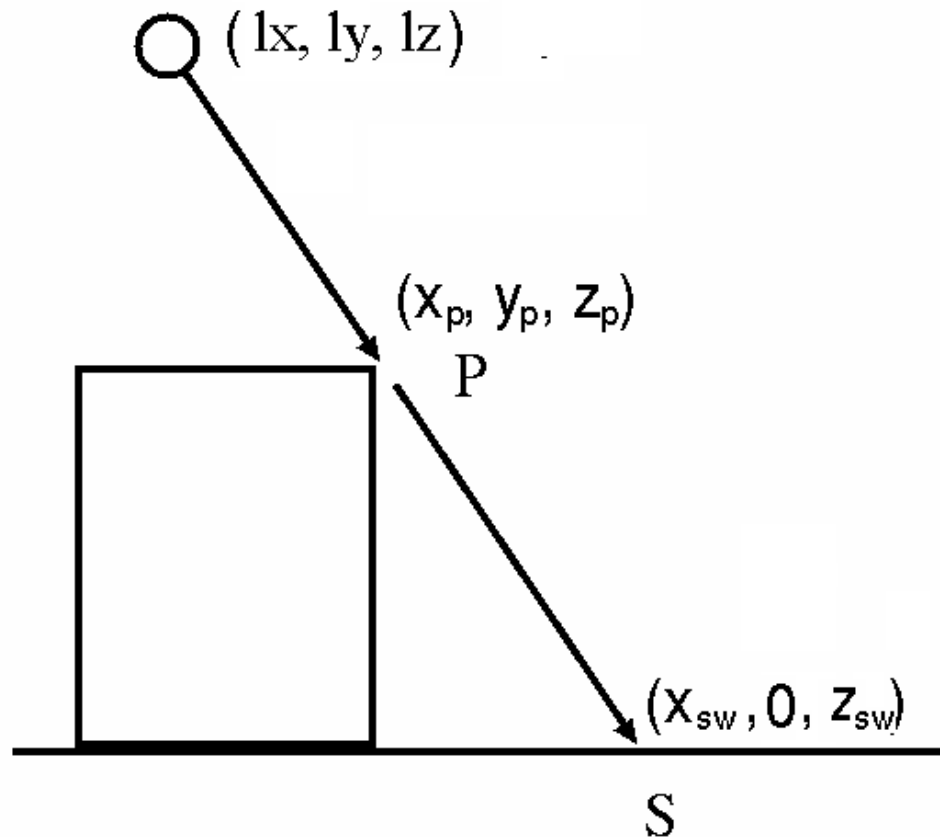
[Tom Thorne, Edinburgh]

Shadow techniques

- Ground shadow
- Shadow texture
- Shadow map
- Shadow volume

Ground shadow

- Shadow cast by objects onto the ground ($y=0$)



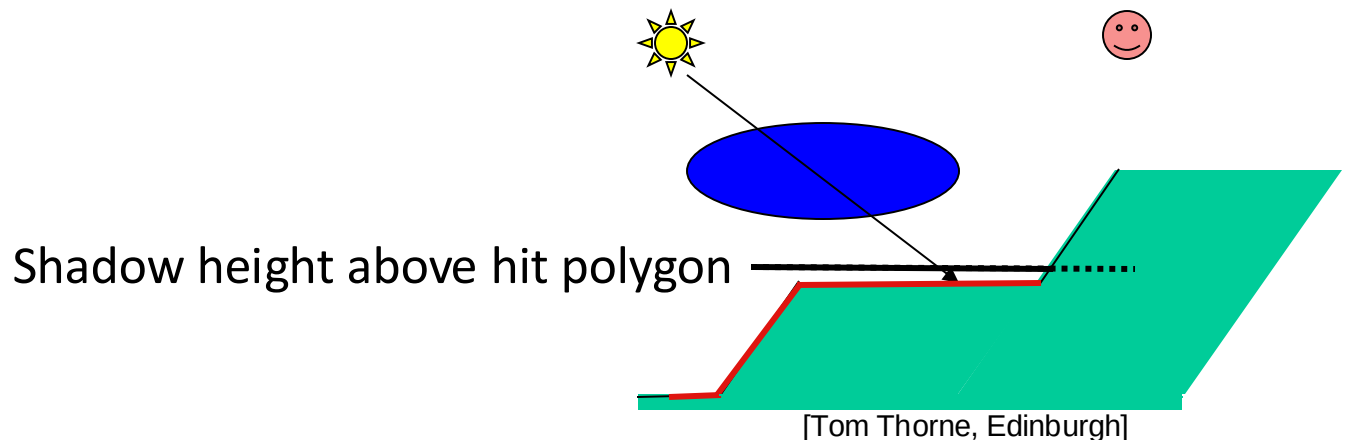
Point light ground shadows

- Blinn 88
- The matrix transform the object onto the ground

$$\begin{bmatrix} X_{sw} \\ 0 \\ Z_{sw} \\ 1 \end{bmatrix} = \begin{bmatrix} l_y & -l_x & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -l_z & l_y & 0 \\ 0 & -1 & 0 & l_y \end{bmatrix} \begin{bmatrix} X_p \\ y_p \\ Z_p \\ 1 \end{bmatrix}$$

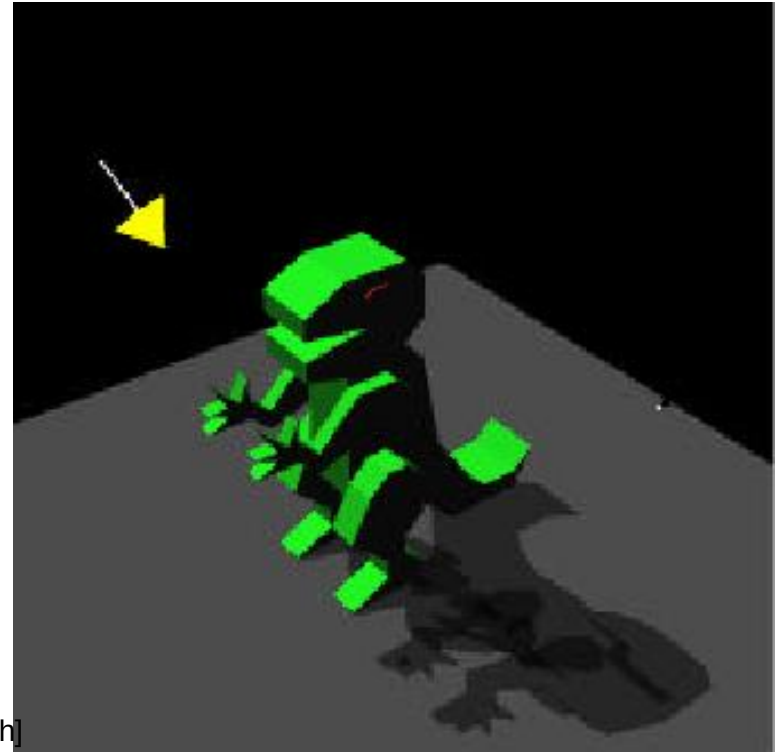
Draw the ground shadow

- The matrix transform the object onto the ground
- Thus
 - Draw the object
 - Multiply the shadow matrix
 - Redraw the object in grey



Problem of ground shadow

- The shadow only cast onto (ground) planes
- The shadow is hard shadow
- The performance is not optimal in static scene



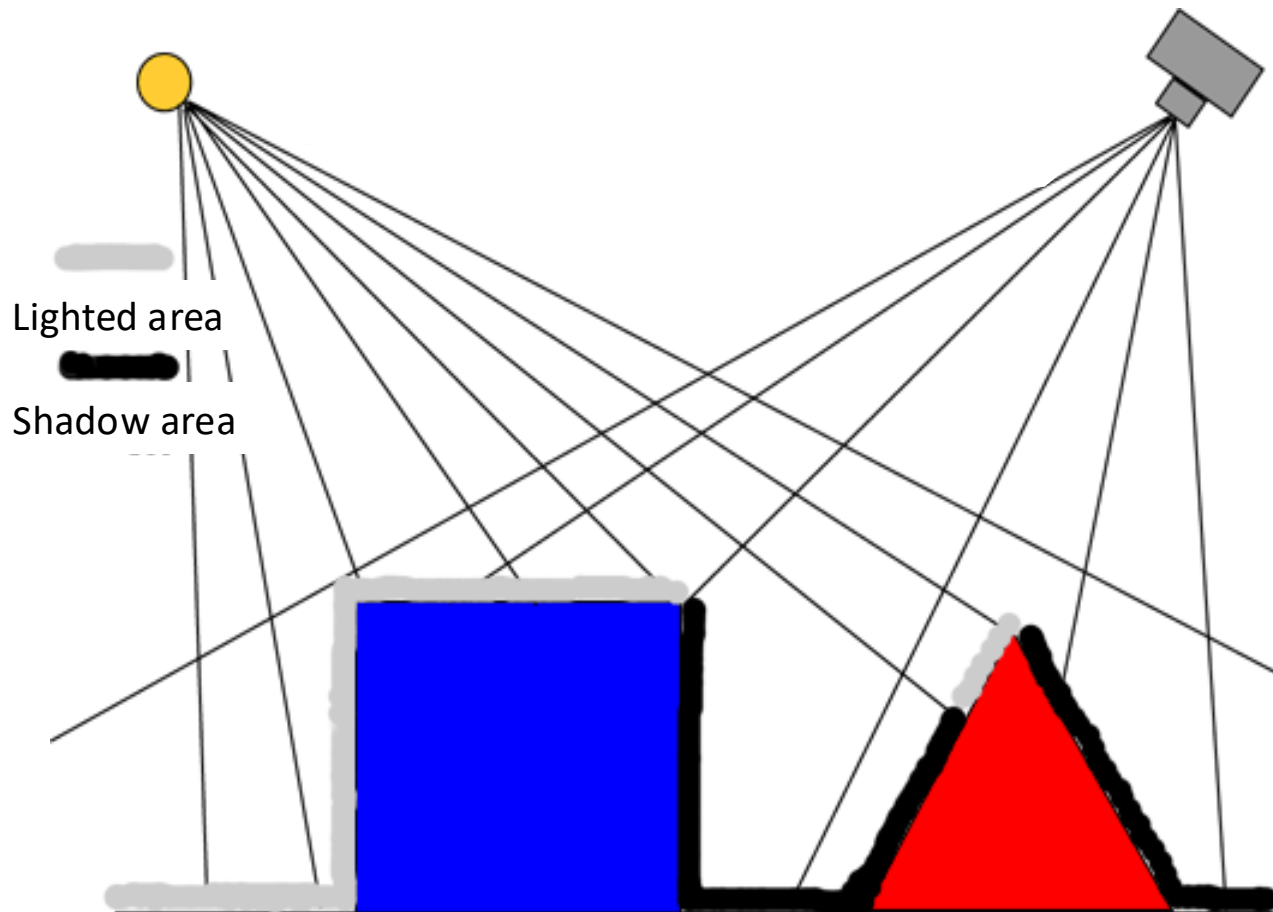
Shadow map

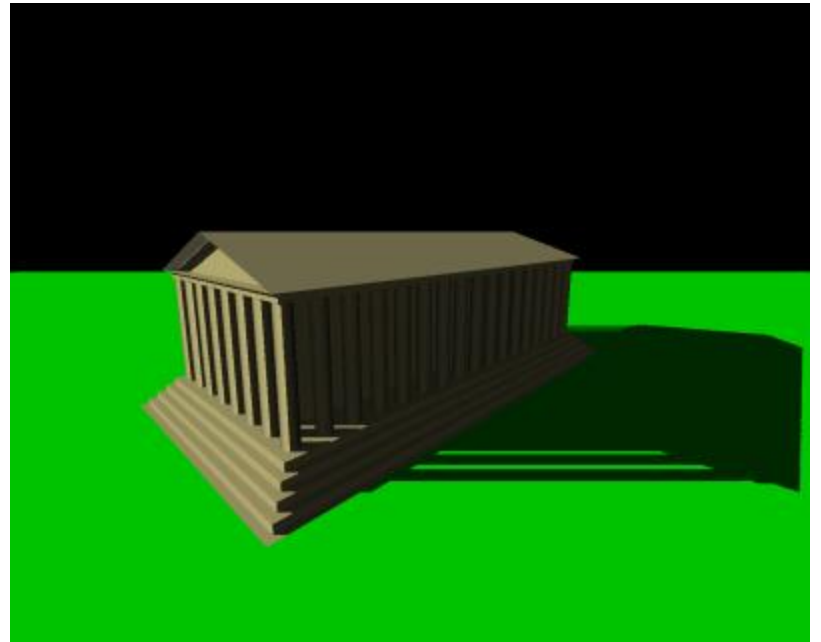
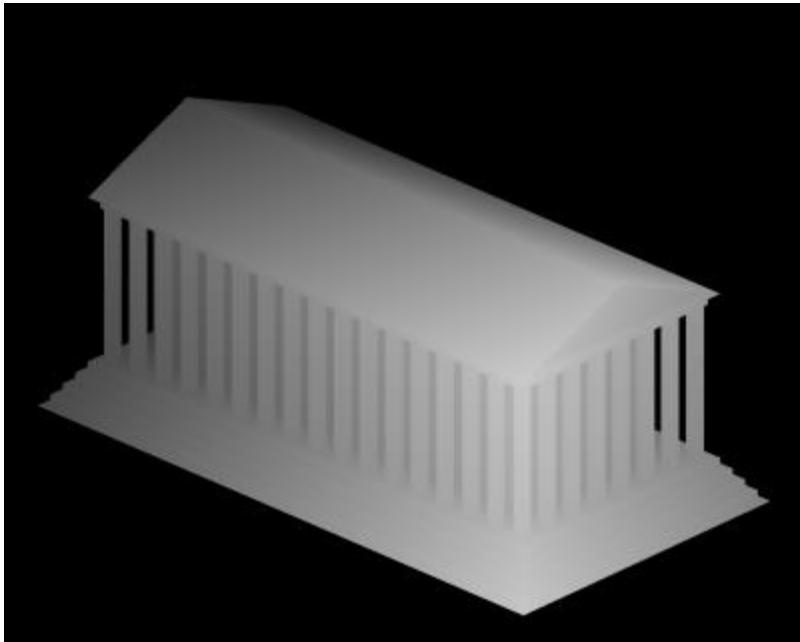
- Preparation
 - Prepare a depth buffer for each light
 - Render the scene from the light position
 - Save the depth information in the Depth buffer
- Rendering the scene
 - Render the objects; whenever rendering an object, check if it is shadowed or not by transforming its coordinate into the light space
 - After the transformation, if the depth value is larger than that in the light's depth buffer it should be shadowed

https://learnopengl.com/code_viewer_gh.php?code=src/5.advanced_lighting/3.1.2.shadow_mapping_base/shadow_mapping_base.cpp

Shadow map

- Using Depth buffer





[Wikipedia]

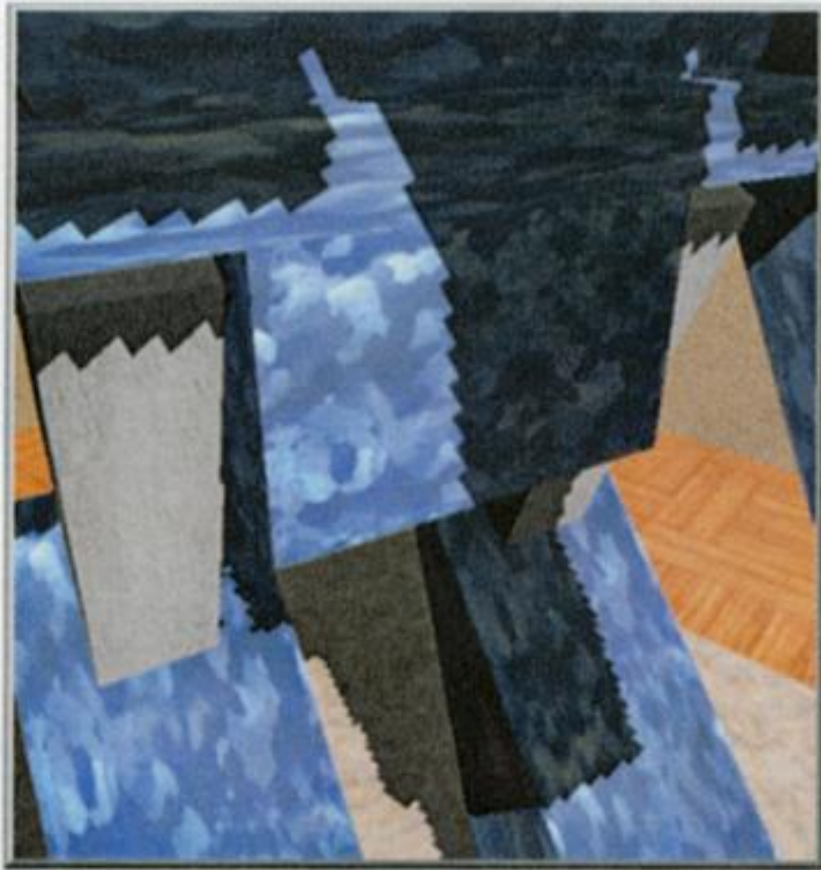
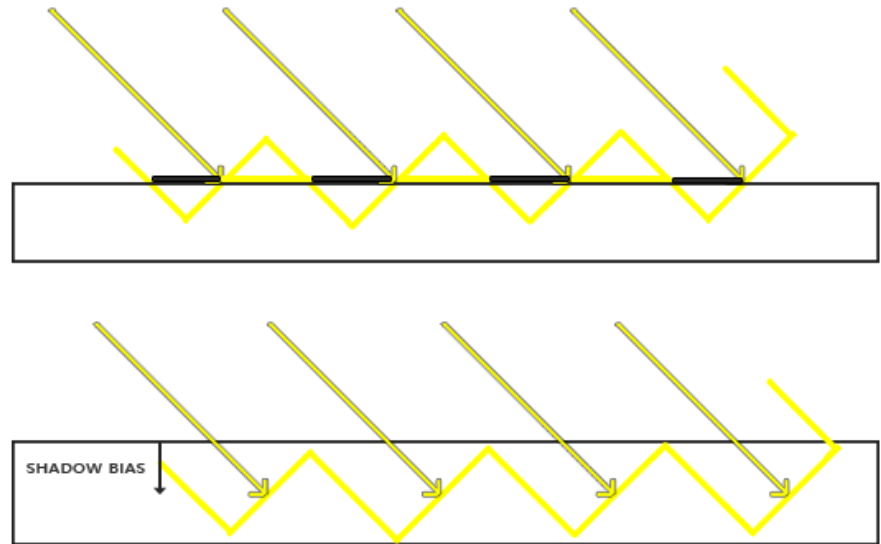
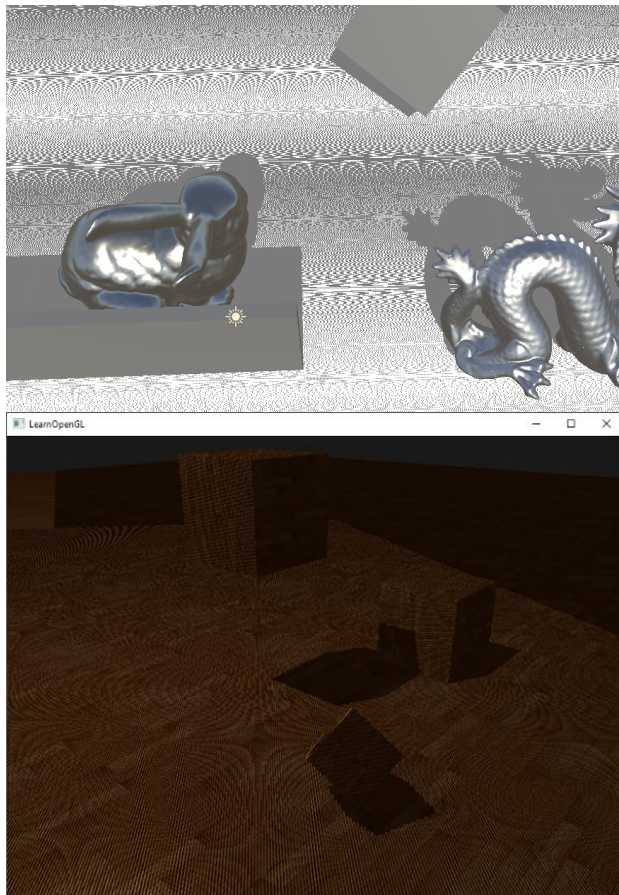


Figure 2: A conventional $2,048 \times 2,048$ pixel shadow map (left) compared to a 16 MB ASM (right).
EFFECTIVE SHADOW MAP SIZE: $65,536 \times 65,536$ PIXELS.

Shadow acne

- Fragment depth



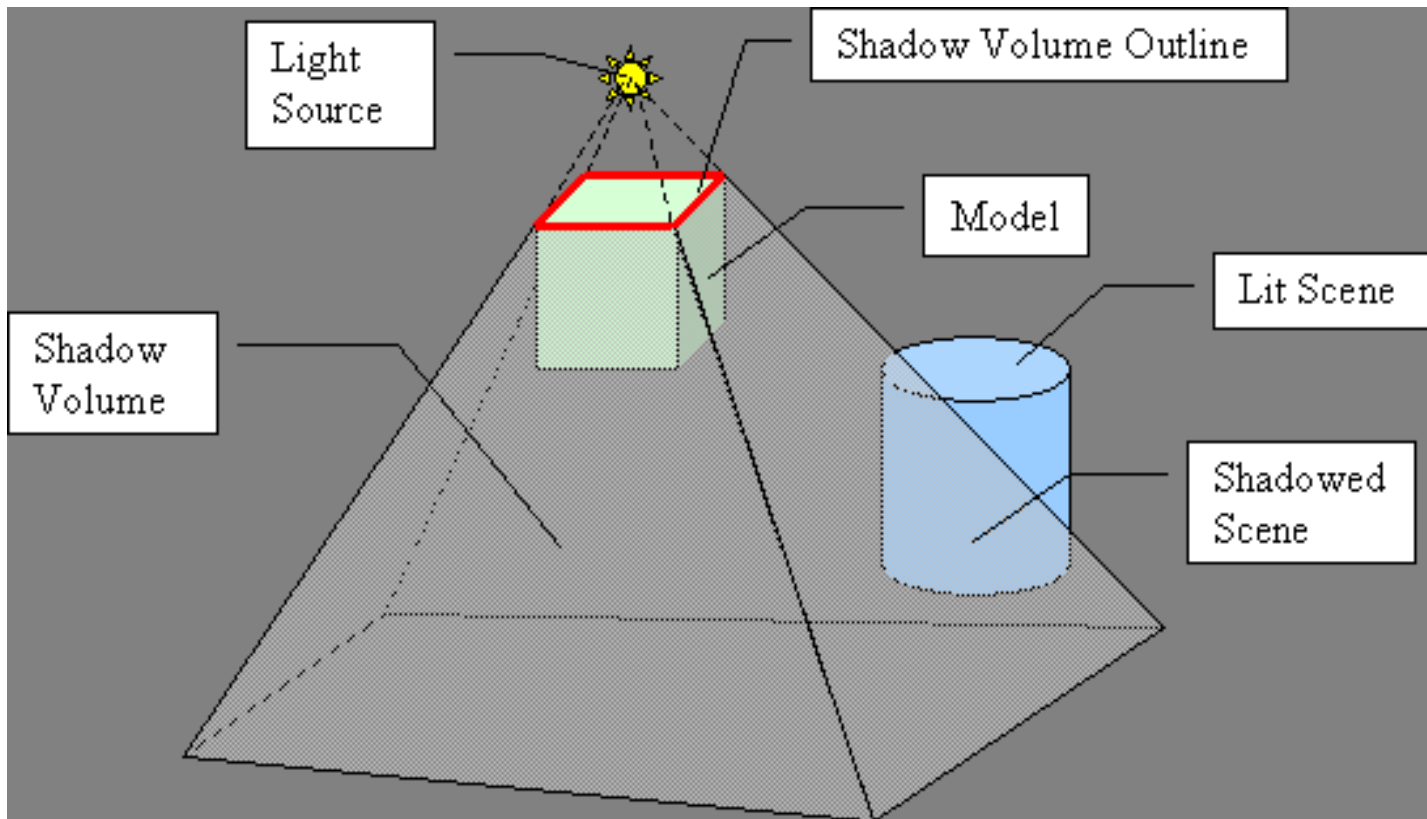
[LearnOpenGL - Shadow Mapping](#)

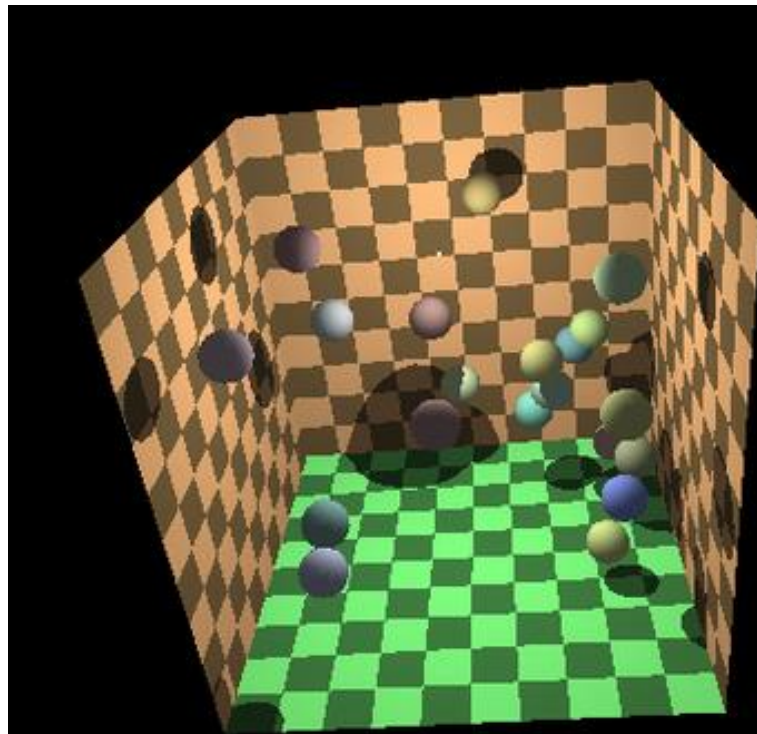
Shadow volume

- In the reality, the shadow cast by an object blocking light is a volume which is 3D!
- Project a ray from the light source through each vertex/silhouette in the shadow caster to infinity
- Any objects intersecting with the volume will get shadow on them.
 - Self-cast shadow
 - General-purpose

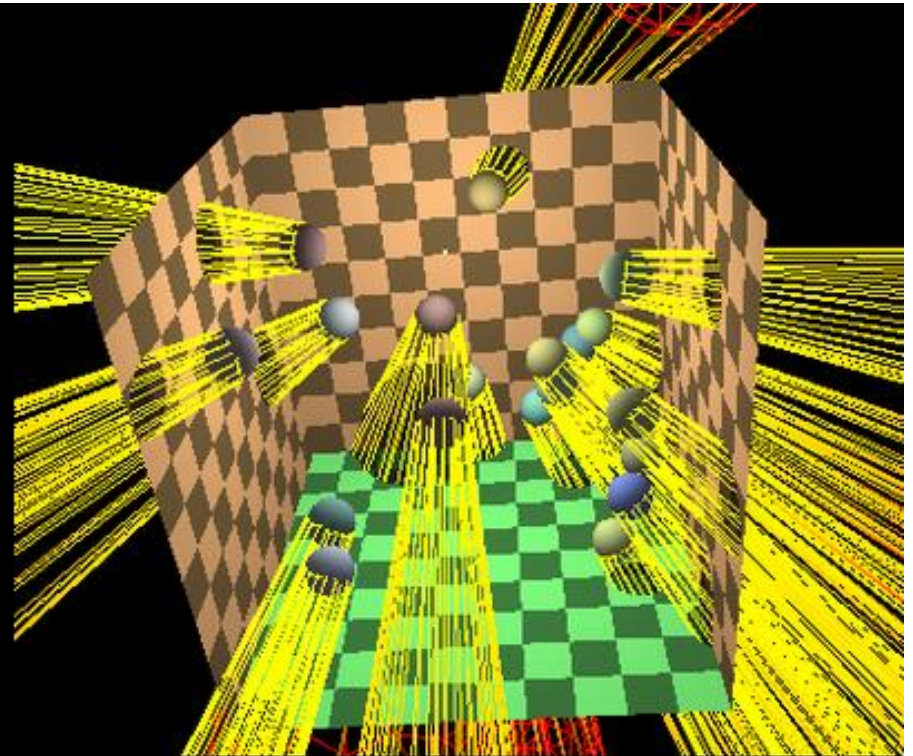
Shadow volumes

- Cast by silhouette





shadowed scene

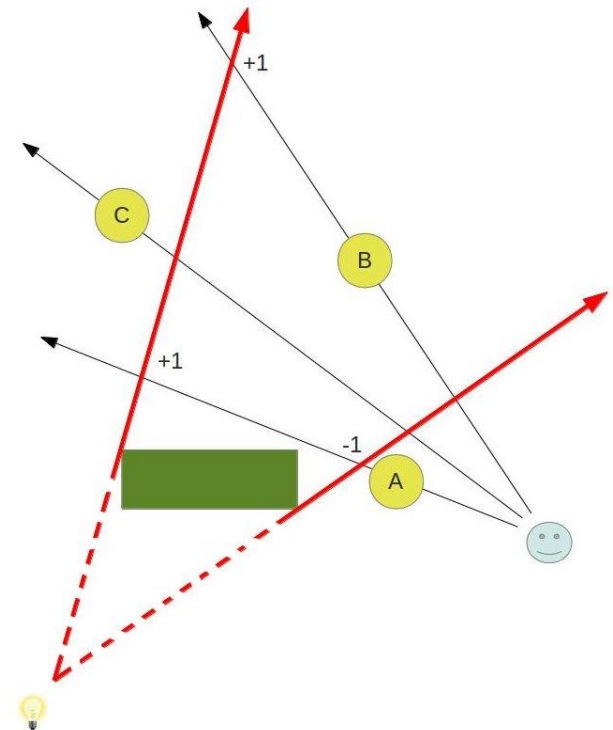


wireframe shadow volumes

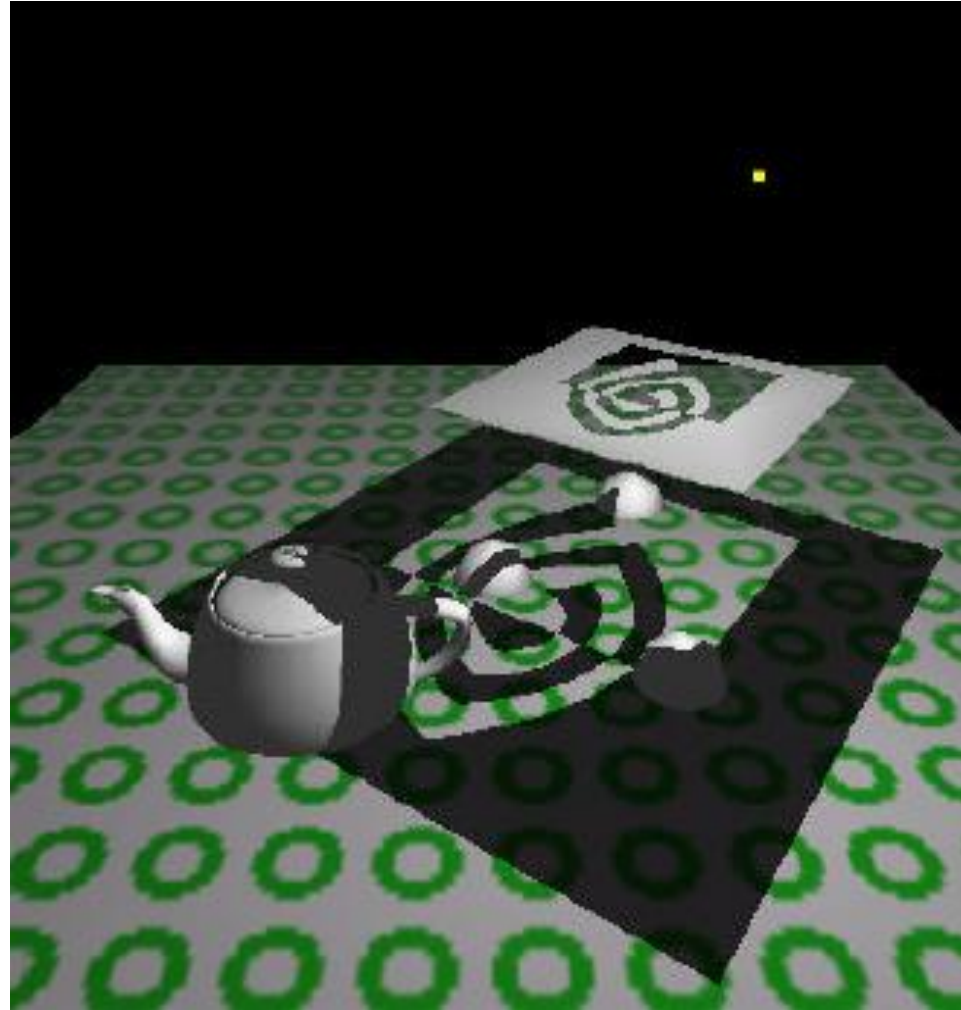
https://en.wikipedia.org/wiki/Shadow_volume

Stencil buffer

- Stencil buffer
 - A data buffer
 - Used as a Stencil
 - An integer per pixel
- Depth fail method
 - <http://ogldev.atspace.co.uk/www/tutorial40/tutorial40.html>
 - 1 Render the scene into Depth buffer
 - 2 Create shadow volume
 - 3 Render the volume in Stencil buffer following rules
 - 4 Render the scene under Stencil test



- Computation intensive
- Hard shadow



Stencil buffer based implementation

- Doom3 Made it popular

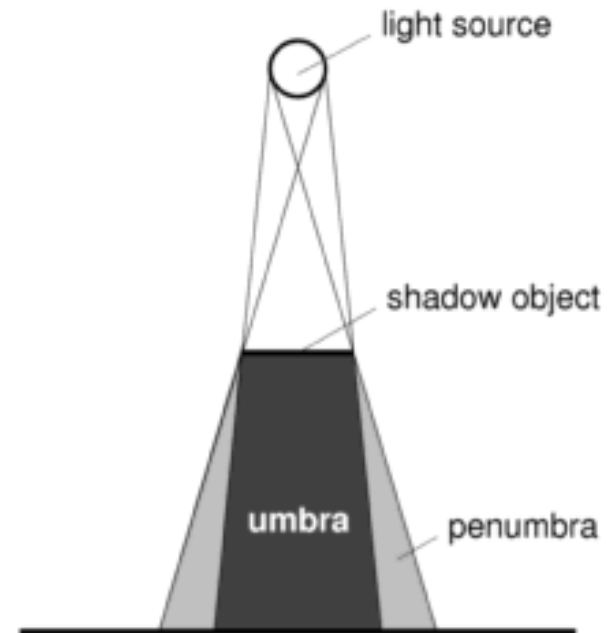


Shadow map vs Shadow volume

- Faster than shadow volume
- Less accurate because the resolution of the depth buffer
 - Aliasing at edges

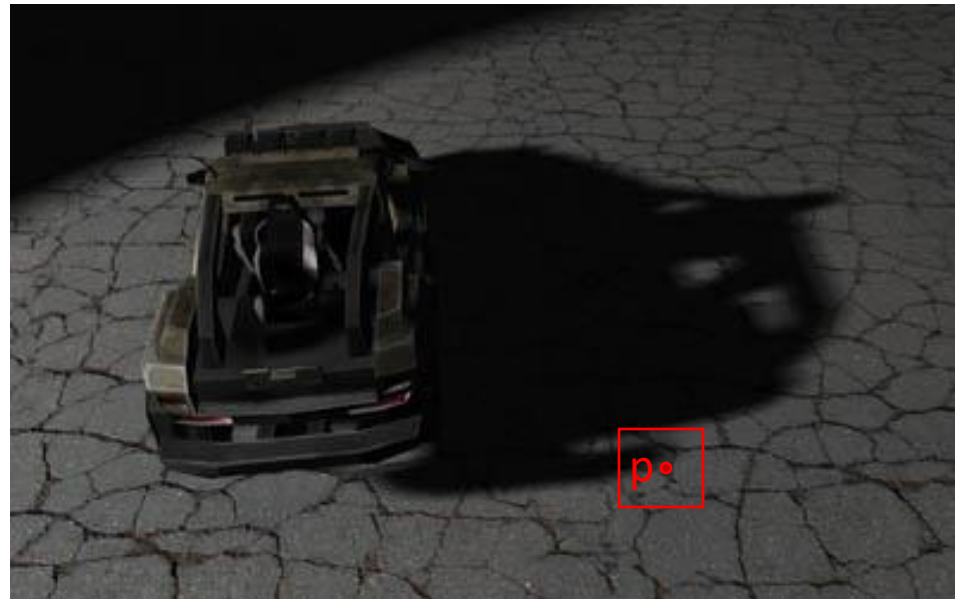
Soft shadows

- Made by area light
 - umbra – totally blocked from the light source
 - Penumbra – partially blocked from the light source
- Can be modelled by a collection of point light sources



PCF (Percentage Closer Filtering)

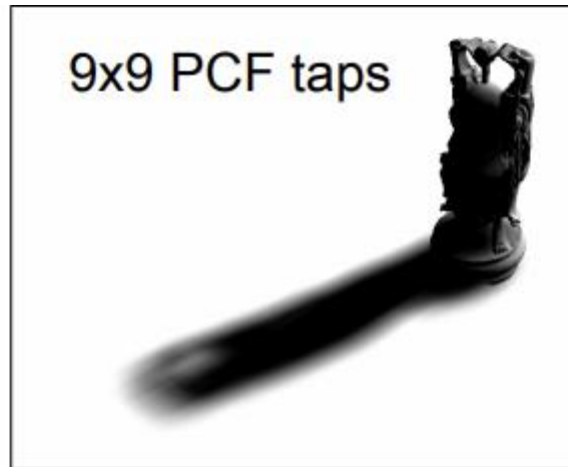
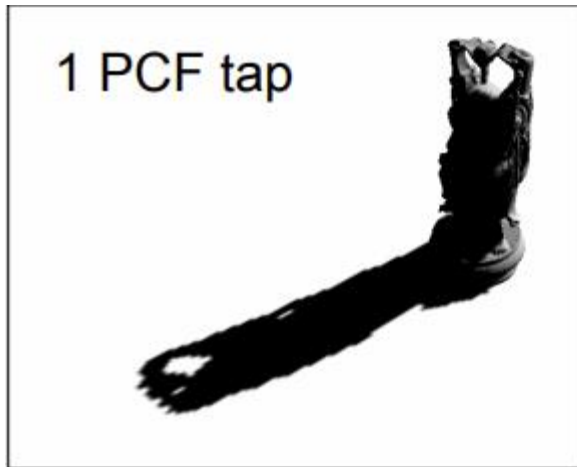
- Anti-aliasing
- [Reeves, SIGGRAPH 87]
 - Perform multiple (e.g. 7x7) depth comparisons for each fragment
 - Then, averages results of comparisons
 - e.g. for point P on the floor,
 - (1) compare its depth with all pixels in the box, e.g. 3x3
 - (2) get the compared results, e.g. 1, 0, 1, 1, 0, 1, 1, 1, 0,
 - (3) take avg. to get visibility, e.g. 0.667



GPUGem3

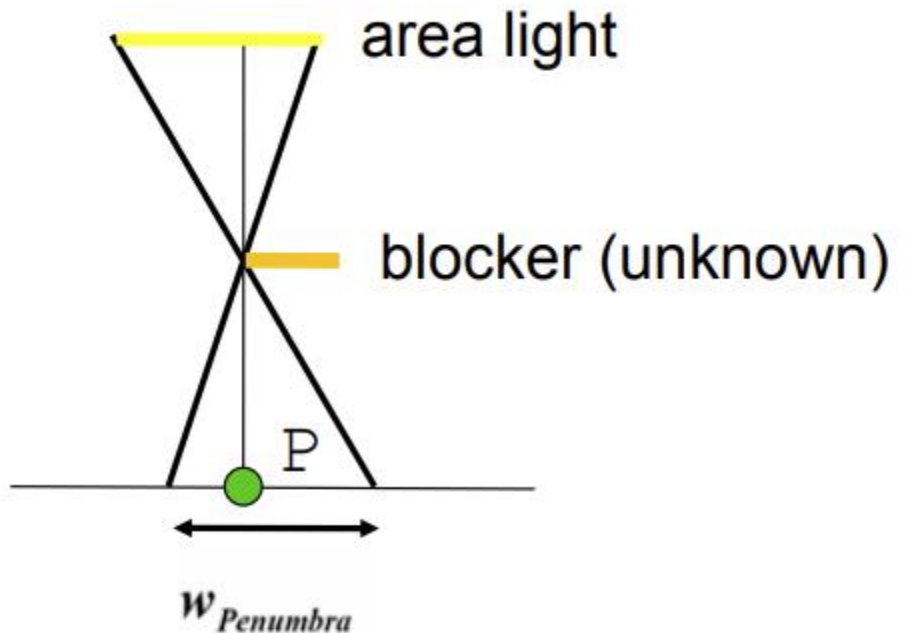
Problems with PCF

- Computational expensive
 - Sampling
- Not real soft shadow



PCSS (Percentage-Closer Soft Shadows)

- PCSS [Fernando05]
- Assume a square light centered at the shadow map center
- Assuming some parallel blocker to receiver
 - Compute penumbra width using similar triangles



$$w_{Penumbra} = \frac{(d_{Receiver} - d_{Blocker}) \cdot w_{Light}}{d_{Blocker}}$$

PCSS vs PCF



[Advanced Soft Shadow Mapping Techniques \(nvidia.com\)](https://nvidia.com/en-us/gameworks/resources/advanced-soft-shadow-mapping-techniques/)

- Questions?

References

- Steve Marschner, CS4620/5620 Computer Graphics, Cornell
- Ed Angel, CS/EECE 433 Computer Graphics, University of New Mexico
- Elif Tosun, Computer Graphics, The University of New York